MIL-STD-1312-2
19 OCTOBER 1984
SUPERSEDING
MIL-STD-1312 (In part)
31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 2,

INTERACTION

FSC 53GP

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 2, Interaction

MIL-STD-1312-2

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard sets forth a standard test method to determine the interaction life cycle capabilities of any configuration of installed fastener when subjected to combined tension and shear loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Applicability	1
2.	REFERENCED DOCUMENTS	1
2.1	Government documents	1
2.2	Other publications	1
3.	DEFINITIONS	1
4.	GENERAL REQUIREMENTS	1
4.1	Test apparatus	1
4.1.1	Test fixtures	1
4.1.2	Instrumentation	1
4.2	Test specimens	1
4.2.1	General	7
5.	DETAIL REQUIREMENTS	7
5.1	Test procedures	7
5.1.1	Installation	7
5.2	Loading rate	7
6.	NOTES	7
6.1	Test report	7
6.1.1	Yield point data	8
6.1.1.1	Autographic curves	8
6.1.1.2	Yield data	8
6.1.1.3	Yield strength	8
6.1.2	Interaction diagram	8

FIGURES

Figure		Page
1.	Assembly, interaction test fixture	2
2.	-1 Side plate, interaction test fixture	3
3.	-2 Retaining block, interaction test fixture	4
4.	-3 Bushing, interaction test fixture	5
5.	Adapter rod and lock pin, interaction test fixture	6
6.	Interaction test diagram	9

TABLE

Table		Page
I.	Load rates	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Applicability. The interaction test for a complete fastener system is a method for determining the properties of the fastener system under combined tension and shear loading for any configuration of installed fastener system.

2. REFERENCED DOCUMENTS

2.1 Government documents.

Not applicable.

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 83 Verification and Classification of Extensometers

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Test fixtures. The fixtures for performing the interaction test on 1/2-inch and smaller fasteners is shown on figures 1 through 5. This fixture shall be adapted by transition studs to any standard type testing machine capable of applying a tensile load smoothly at a controlled rate. Test bushings (figure 4) shall be modified to suit the configuration of the fastener to be tested. Fixtures for larger fasteners shall be proportionately larger, similar in shape and design, and equivalent in dimensional control.

4.1.2 Instrumentation. The fixture shown on figure 1 provides a means of attaching an extensometer with 7-inch gage length. The extensometer shall be an averaging, differential-transformer extensometer, or equivalent. It shall conform to ASTM E 83, class B-1.

4.2 Test specimens.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

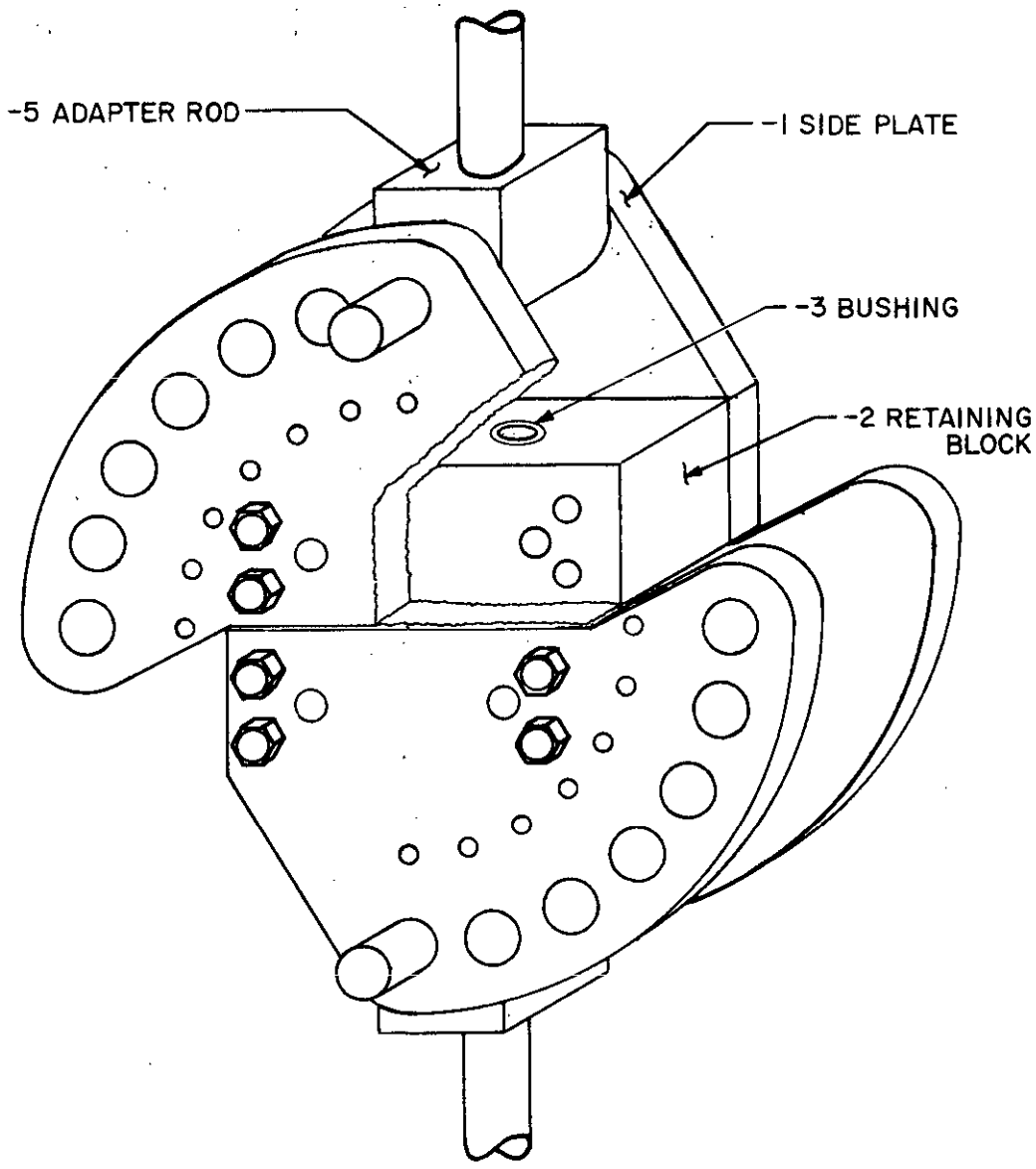
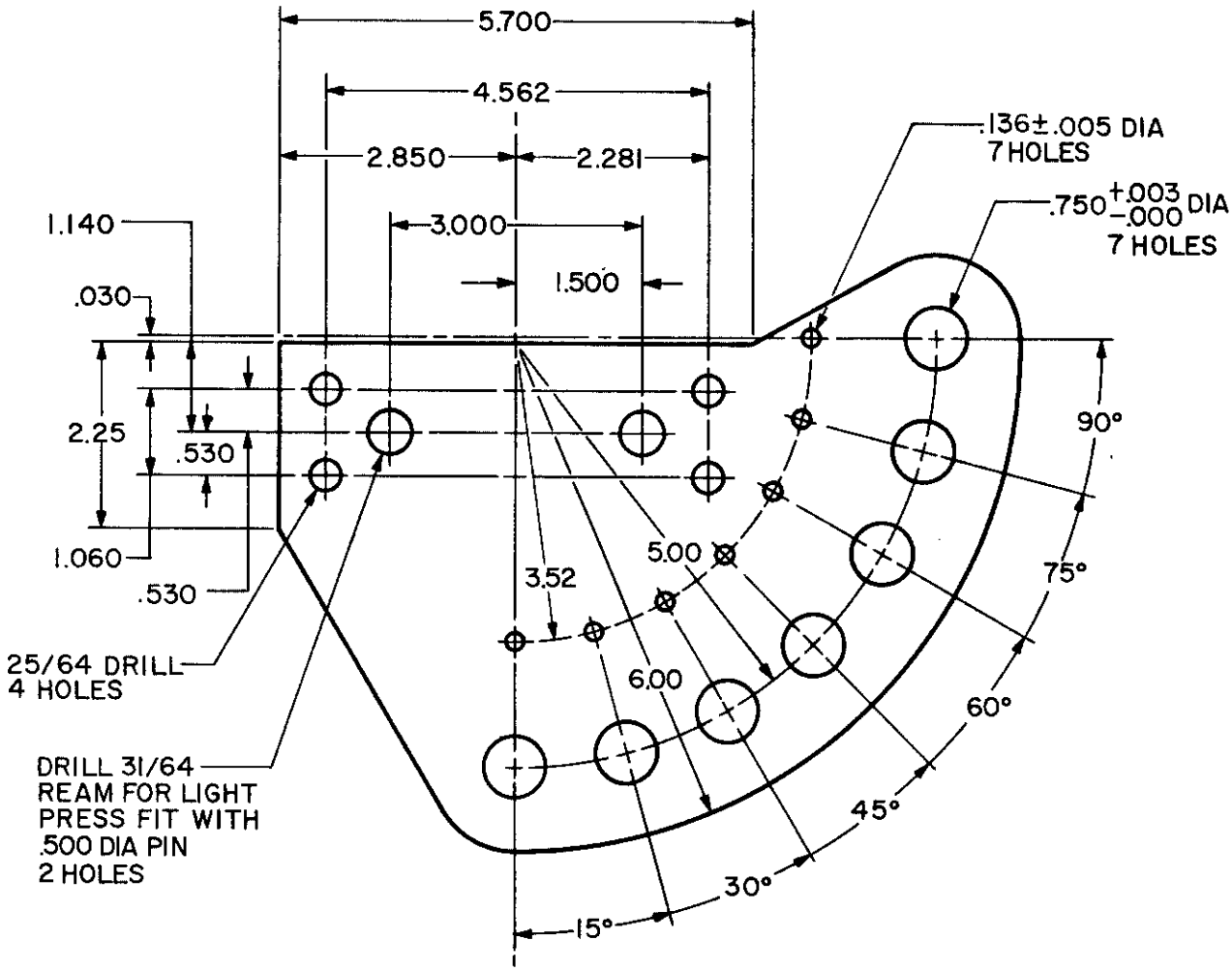


FIGURE 1. Assembly, interaction test fixture.

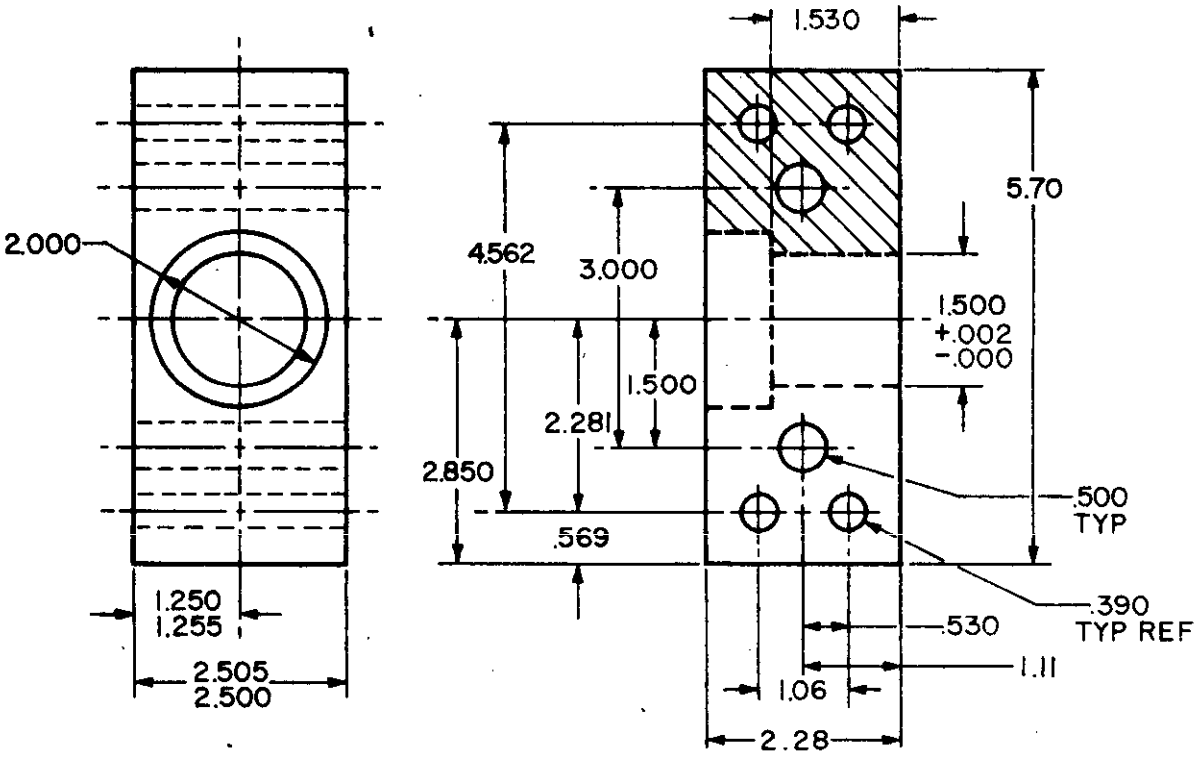
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.



NOTE:
1. All dimensions in inches.
2. Unless otherwise specified tolerance ±.030 for .XXX.
   ±.010 for .XX.
3. Material thickness .500 inch.

FIGURE 2. -1 Side plate, interaction test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

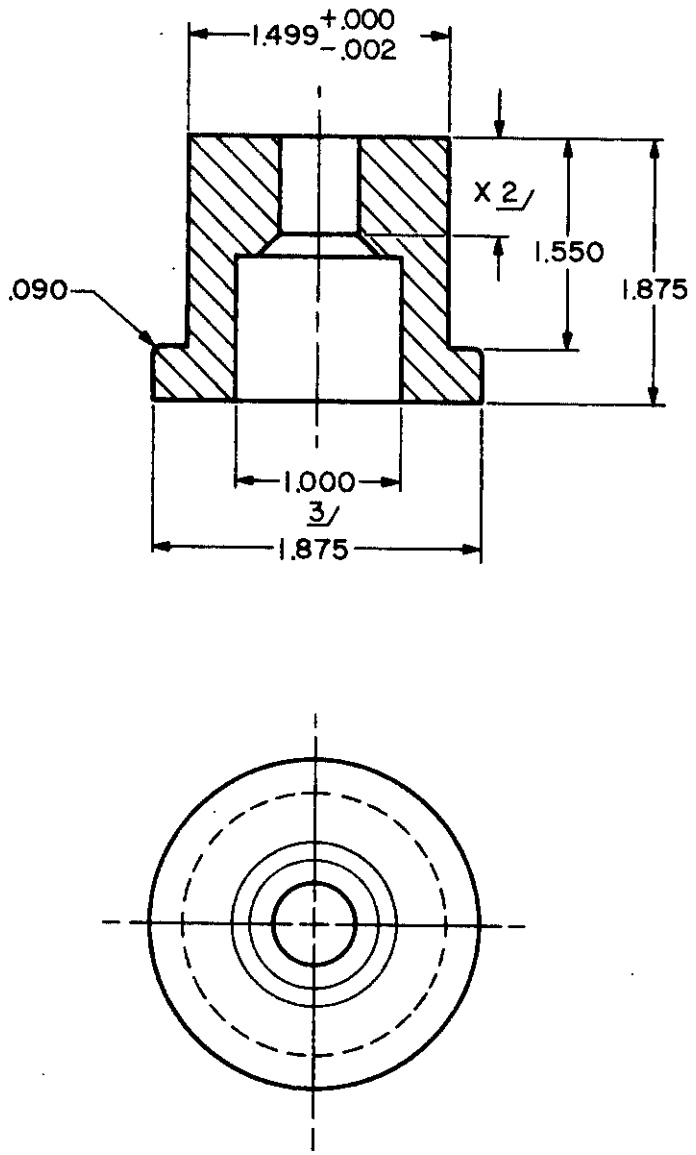


NOTES:
1. Unless otherwise specified, tolerance: Decimals ± 0.010
   Angles ± 0° 10'

2. All dimensions in inches.

FIGURE 3. -2 Retaining block, interaction test fixture.

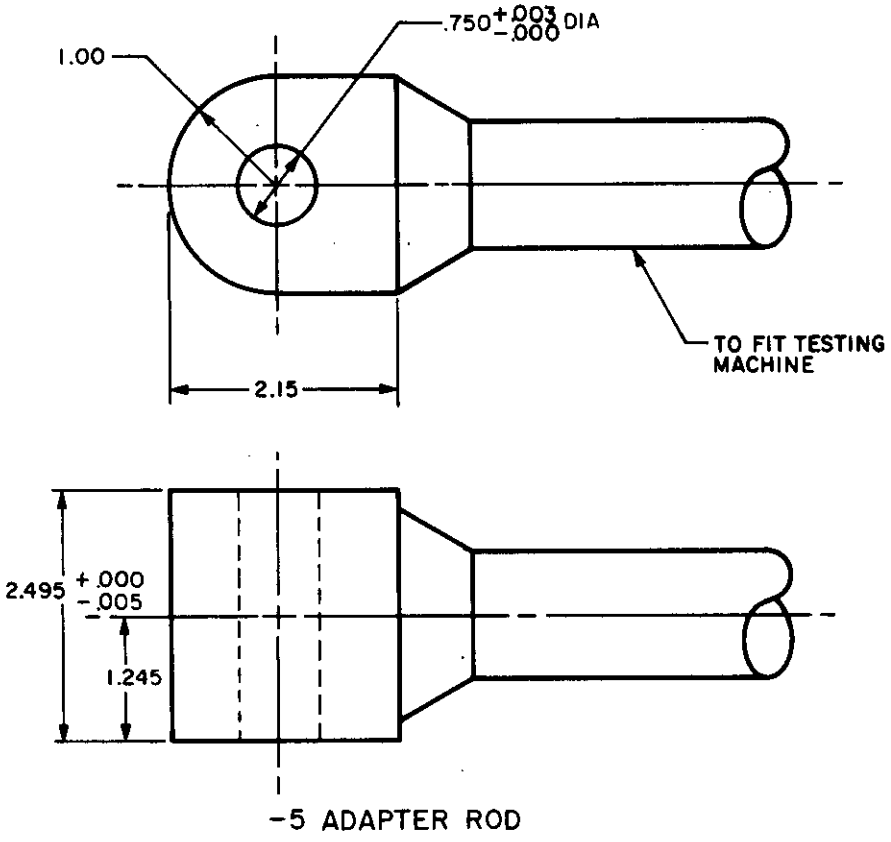
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.



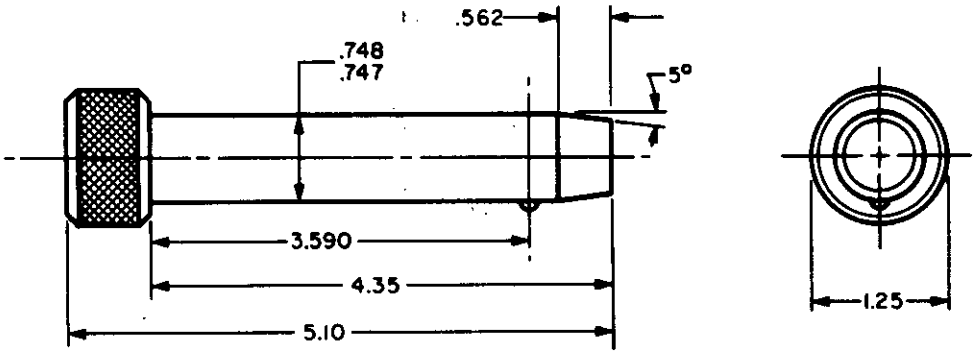
NOTES:
1. Unless otherwise specified, tolerances: Decimals ± 0.010
   Angles ± 0° 10'
2. X to be 2 times diameter of part being tested.
3. Hole diameter and countersink in accordance with part tested.
4. All dimensions in inches.

FIGURE 4. -3 Bushing, interaction test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.



-5 ADAPTER ROD



-4 LOCK PIN

NOTES:
1. Unless otherwise specified, tolerances: Decimals ± 0.010
   Angles ± 0°
2. All dimensions in inches.

FIGURE 5. Adapter rod and lock pin, interaction test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

4.2.1 General. This test evaluates a complete assembly. Tests resulting in failure of any component of the assembly shall be considered a complete test of that fastener. When testing either an internally- or externally-threaded fastener, the mating part shall be of sufficient strength to ensure failure in the test specimen. The grip length shall be at least four times the nominal diameter.

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Installation. The fastener shall be installed in the fixture with standard production tools, using the manufacturer's recommended installation procedures.

5.2 Loading rate. Tension loads shall be applied slowly and evenly; unless otherwise specified, the rate of loading shall be in accordance with table I.

TABLE I. Load rates.

Nominal size (in.)	Load rate ±10% (lb/min)	Nominal size (in.)	Load rate ±10% (lb/min)
0.125	1,240	0.563	24,800
0.156	1,920	0.625	30,600
0.164	2,100	0.750	44,000
0.190	2,800	0.875	60,000
0.250	5,000	1.000	78,000
0.313	7,700	1.125	100,000
0.375	11,000	1.250	122,000
0.438	15,000	1.375	148,000
0.500	19,600	1.500	176,000

6. NOTES

6.1 Test report. The test report shall include the following data:

a. Fastener type and part number, including lot identification, fastener diameter, and description of material(s).

b. Testing machine used with most recent date of certification and calibration, and method used for calibration.

c. Interaction angle of test.

d. Individual readings for ultimate strength achieved for each group or specimen.

e. Computation of average ultimate strength for each specimen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

f. Extensometer and gage length.

g. Installation procedure.

h. Type of failure.

i. Cause and duration of any interruption during test.

j. Results of all inspections.

6.1.1 Yield point data (optional).

6.1.1.1 Autographic curves. Autographic curves for each specimen shall be included with the test data when yield point information is required. Each curve shall show the graphic determination for yield point for each specimen, and shall be completely identified as to specimen, test number, and date of test.

6.1.1.2 Yield data. Yield data shall be presented on the data tabulation sheet for each specimen.

6.1.1.3 Yield strength. Yield strength shall be determined as specified in the procurement document.

6.1.2 Interaction diagram.

6.1.2.1 An interaction diagram may be plotted, using applied load laid out to scale on the angle of the test. The tension load in pounds is the ordinate and the shear load in pounds is the abscissa. Units of the ordinate and abscissa must be equal in value and scale. (Pure tension is on the fastener at 0-degree interaction angle and pure shear load is on the fastener at 90-degree interaction angle.) A sample diagram is included as figure 6.

6.1.2.2 A smooth curve may be drawn between plotted points and intermediate loads picked off this curve. A curve should be drawn for the average load of all tests, the low load recorded for all tests, and when desired, the yield load curve may be drawn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

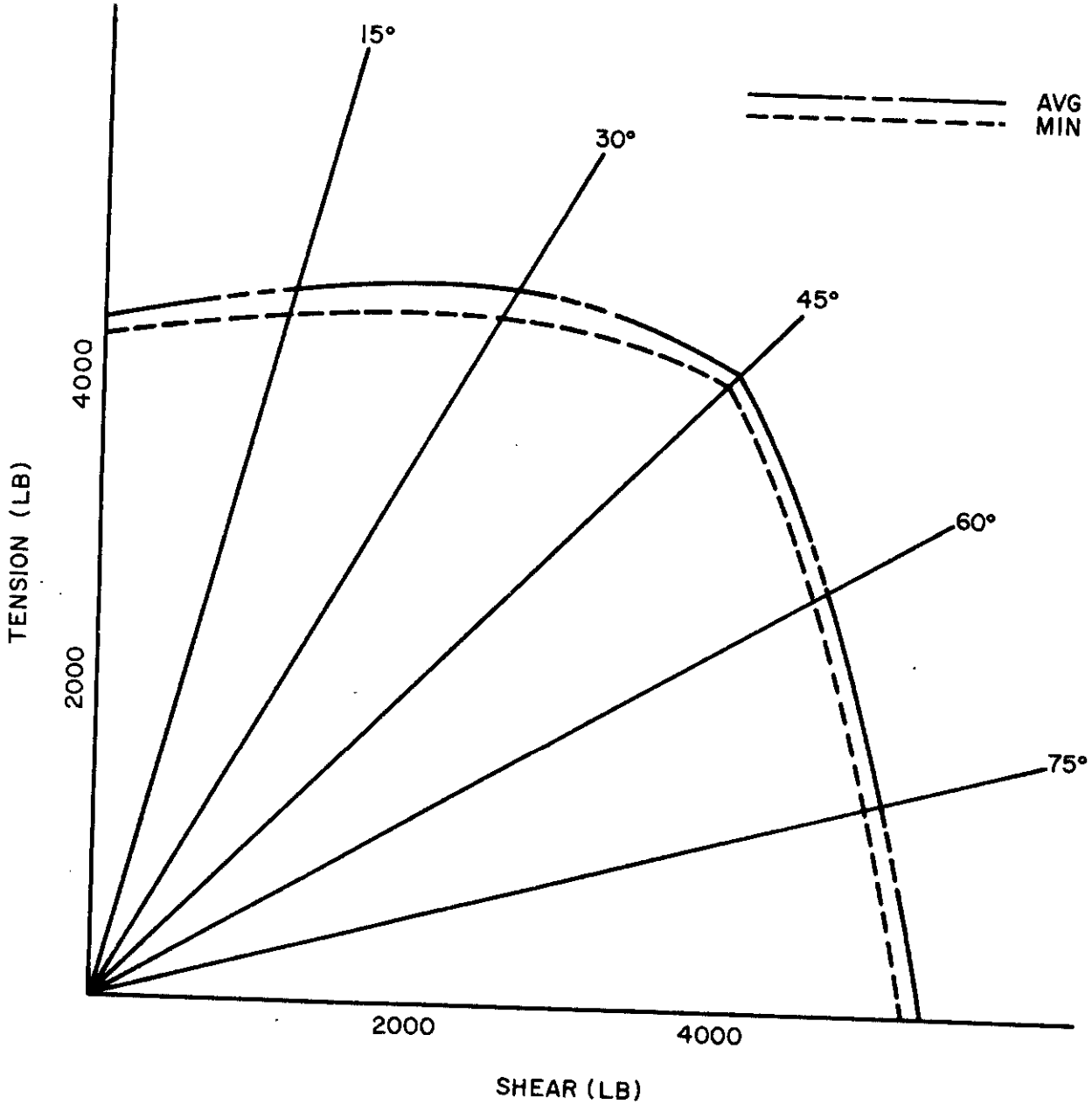


FIGURE 6. Interaction test diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army - AV, AR
Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0143-02)

☆ U.S. GOVERNMENT PRINTING OFFICE: 1984-505-038/A5113

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1312-2	Fastener Test Method - Method 2, Interaction

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.